Predictive Deadlock Warning — Project Report

Operating Systems case study, CBIT (Information Technology)

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1. Summary in one page

Question. Banker's algorithm tells an operating system whether the current state is *safe right now*. It does not warn that the system is *heading towards* danger. We asked: **can simple, explainable signals raise an alarm a few steps before Banker's algorithm reports an unsafe state — how early, and how reliably?**

What we built. A resource-allocation simulator (with Banker's safety check and deadlock detection), three workload scenarios, a rule-based early-warning algorithm (no machine learning), an evaluation method borrowed from hospital and earthquake early-warning systems, and an experiment pipeline. Everything is tested by 122 automated tests.

How big the experiment was. 195,000 simulated sequences — 440 million resource requests, each followed by a Banker's safety check — across 3 scenarios and 5 system sizes (10 to 200 processes), run on AWS EC2 **c7i.8xlarge** (32 vCPUs, 61.8 GB RAM, region ap-south-2 / Hyderabad). Settings were tuned on 45,000 sequences and all reported numbers come from a separate 150,000 sequences the tuner never saw.

Main answers.

1. **Yes, early warning works — moderately.** Our combined score catches 34% of unsafe onsets in advance with 24% of its alarms correct (F1 = 0.28), giving a median **10 steps of warning**. Random alarms at the same rate score F1 = 0.01, and the simple "free resources are low" rule scores 0.16.
2. **Contention is the single most useful signal** (F1 = 0.34 on its own). Averaging in weaker signals dilutes it: the tuner's best combination doubled contention's weight ("double_contention").
3. **Machine learning on the same cheap signals helps a lot.** Gradient-boosted trees reached F1 = 0.48 (vs 0.28 for our rule), while logistic regression gained little (0.31): the signals matter *in combination*, in ways a simple weighted sum cannot capture. The trade-off is explainability.
4. **Checking the safety boundary directly is most accurate, and does not have to be expensive.** On the runs where every method was computed, exact lookahead scored 0.74 but costs 43× a whole simulation step at 200 processes; a *safety margin* (how much free resource could disappear before the state turns unsafe) scored 0.72 for only about 1.7× the cost. Both warn later than the cheap signals.
5. **Sudden bursts are easiest to predict, slow build-ups hardest** (F1 0.42 for spikes vs 0.22 for normal workloads).
6. **"Unsafe" almost always became "deadlock"** in our model (100% of unsafe episodes for spikes, 77% for normal workloads), because processes really do end up requesting their full declared claims.

2. Background: the ideas in plain words

- **Process:** a running program that holds and requests resources.
- **Resource type:** a kind of resource (memory blocks, a printer, a file lock) with a fixed number of units.
- **Maximum claim:** the most units of each type a process says it may ever need. **Need** = claim − what it holds.
- **Safe state:** Banker's algorithm can find *some order* in which every process could get its full need and finish.
- **Unsafe state:** no such order exists. This is *danger*, not yet necessarily a deadlock.
- **Deadlock:** some waiting processes can never be satisfied even if everyone else finishes. A deadlocked state is always unsafe; a safe state is never deadlocked (our tests check this on thousands of random states).

Banker's algorithm is like a fire alarm: it rings when the fire is already there. We are testing a smoke detector.

3. What we built

3.1 The simulator ("observer mode")

In a real system, Banker's algorithm *refuses* dangerous requests, so the system never becomes unsafe — and there would be nothing to predict. Our simulator therefore **grants every valid request** and uses Banker's algorithm only to *watch*. Every step:

1. Waiting processes are retried, longest-waiting first; a process whose claim is now fully met finishes and returns everything, and the retry repeats with the freed units.
2. One process makes a new request (never zero, never more than its remaining need). If the units are free it is granted — **even if this makes the state unsafe** — otherwise the process waits.
3. Finished processes release their resources; in the *normal* scenario a process may also give some back early.
4. We record: is the state unsafe (Banker's)? is it deadlocked (detection)? and the five warning signals.

A run ends at the first deadlock, when every process has finished, or at a step limit.

3.2 Three workloads

Scenario	What happens	What it tests
normal	small requests, frequent completions, occasional early releases	false alarms; slow build-up
spike	normal traffic, then 80% of processes suddenly grab large chunks and hold them	sudden change
deadlock_prone	process *i* keeps asking for resource *i+1* (a circular pattern) and rarely finishes	gradual build-up



Figure 1: Example test runs, one per scenario: the risk score (blue) rises, an alarm fires (triangle), then the system turns unsafe (red band) and deadlocks (red line).

The key setting is **claim tightness** (total declared claims ÷ total units). We calibrated it on practice seeds so each scenario gives a mix of runs that do and do not become unsafe (Section 6.1).

3.3 The warning algorithm

After every step we compute five numbers between 0 and 1:

Signal	Meaning
Scarcity	how much of all resources is already handed out
Need pressure	how much is still wanted compared with what is free
Contention	for how many resource types two or more processes want more than is free
Waiting ratio	what fraction of processes are blocked
Request rate	whether requests are speeding up compared with the previous 10 steps

They are combined into a **risk score from 0 to 100**. An alarm fires when the score stays above a threshold for k steps in a row, and then stays quiet for R steps so that one long risky stretch counts once. The tuned setting was threshold **65**, k = 1, R = 10, using weights "double_contention".

4. Every edge case we handle

Each item below has an automated test (122 tests in total).

- **States:** safe-and-deadlocked never happens; unsafe-but-not-deadlocked is recorded separately; a starting state that is unsafe is redrawn; all-finished counts as safe and ends the run; units are never created or lost; a finished process never holds anything; the fast Banker's check always agrees with the textbook version.
- **Requests:** never zero; never above the remaining need (flagged and skipped if it ever were); above the free units → wait; a request that makes the state unsafe is still granted (observer mode); waiting or finished processes never request; waiting processes are retried first-come first-served; no possible requester ends the run.
- **Life-cycle:** a process whose need reaches zero finishes and releases everything; unsafe can recover to safe, so one run can have several danger episodes; the same seed always gives the same run; a process finishing during the retry frees units for the next waiter in the same step (a bug we found and fixed).
- **Signals:** all zero when nothing is active; no division by zero when nothing is free; request rate is zero until there is enough history; every signal always stays within 0–1.
- **Alarms:** a score hovering at the threshold needs k steps in a row; a long high stretch fires once per quiet period; no alarms while the system is already unsafe.
- **Scoring:** an alarm 1–20 steps before danger is a true alarm; an alarm at the same step is a *miss* (detection, not prediction); several alarms before one event — the earliest gives the lead time; one alarm cannot claim two events; alarms near the end of a run cut off by the step limit are not counted (the future is unknown), but near the end of a run that ended in deadlock they count as false (the future is known); danger that flickers unsafe→safe→unsafe within 5 steps is one episode; danger in the first 3 steps is excluded for every method.
- **Experiments:** practice (tuning) seeds and test seeds can never mix (checked in code); a crashed run resumes; re-running the same configuration is skipped.

5. How the experiments were run

1. **Dry run on the MacBook** (48,000 sequences) to check the whole pipeline and measure timings.
2. **Official run on AWS EC2**, the same code and the same command (`run_all.sh`), at larger scale:

- **Instance:** 1 x c7i.8xlarge (32 vCPU, 61 GB RAM), Ubuntu 24.04, region ap-south-2 (Hyderabad)
- **Launched (UTC):** 2026-10-21 05:41
- **Terminated (UTC):** 2026-10-21 11:31
- **Instance hours:** 5.84
- **Pipeline wall time:** 5 h 40 min (incl. a 42-min stall from thread oversubscription, fixed and resumed)

- **Estimated cost:** about $8.39 (compute $8.33 at $1.428/h, 40 GB gp3 disk $0.03, public IPv4 $0.03); paid from AWS credits
- **Code:** commit 328caf6; tuning, evaluation and figures re-run with the learned-score NaN fix of commit 173a323
- **Cost guards:** 10-hour OS stop timer; self-stop 45 min after the pipeline; idle-CPU (<2% for 30 min) stop alarm; SSH from one IP only; all resources tagged and deleted
- **Clean-up audit:** no instances, volumes, Elastic IPs, snapshots, security group, alarm or key pair left (deployment/aws_audit.sh: AUDIT CLEAN)

The pipeline: tests → smoke run → calibration → 45,000 practice runs → 150,000 test runs → tuning (≈2,000 settings tried per method) → evaluation with 95% confidence intervals → speed study → tables and figures. Every run stores its configuration, code version (commit `328caf6`) and machine details.

6. Results

6.1 Setup per cell (test split)

Scenario	Processes	Resource types	Sequences	Steps	With ≥1 unsafe event	Ended in deadlock	Claim tightness
deadlock_prone	100	20	10000	29793804	93%	93%	1.48
deadlock_prone	10	5	10000	1020230	41%	41%	1.48
deadlock_prone	200	20	10000	59959348	93%	93%	1.48
deadlock_prone	25	8	10000	4036769	40%	40%	1.48
deadlock_prone	50	10	10000	9877410	45%	45%	1.48
normal	100	20	10000	39280274	66%	63%	1.43
normal	10	5	10000	1071547	27%	25%	1.43
normal	200	20	10000	82123372	56%	54%	1.43
normal	25	8	10000	4494155	16%	14%	1.43
normal	50	10	10000	11527074	11%	10%	1.43
spike	100	20	10000	27023589	62%	62%	1.58
spike	10	5	10000	707662	71%	71%	1.58
spike	200	20	10000	56973051	53%	53%	1.58
spike	25	8	10000	2939715	60%	60%	1.58
spike	50	10	10000	7434425	53%	53%	1.58

6.2 Main result — can we warn before "unsafe"? (warning window: 20 steps)

Method	Precision	Recall	F1 [95% CI]	False alarms / 1000 steps	Median lead (steps)
Combined score (ours)	0.24	0.34	0.28 [0.28–0.28]	0.33	10
Contention	0.30	0.39	0.34 [0.34–0.34]	0.22	7
Need pressure	0.31	0.14	0.19 [0.19–0.19]	0.07	7
Waiting ratio	0.24	0.26	0.25 [0.25–0.25]	0.31	13
Request rate	0.16	0.12	0.14 [0.13–0.14]	0.15	9
Availability (free-resource rule)	0.11	0.34	0.16 [0.16–0.16]	0.71	8
Logistic regression (ML)	0.24	0.44	0.31 [0.31–0.31]	0.48	12
Gradient-boosted trees (ML)	0.41	0.58	0.48 [0.47–0.48]	0.29	11
Safety margin	0.64	0.61	0.62 [0.62–0.63]	0.08	4
Top-3 lookahead	0.54	0.62	0.58 [0.57–0.58]	0.13	5

Method	Precision	Recall	F1 [95% CI]	False alarms / 1000 steps	Median lead (steps)
Exact lookahead (expensive)	0.69	0.81	0.74 [0.74–0.75]	0.62	4
Random alarms (chance)	0.00	0.01	0.01 [0.01–0.01]	0.44	11

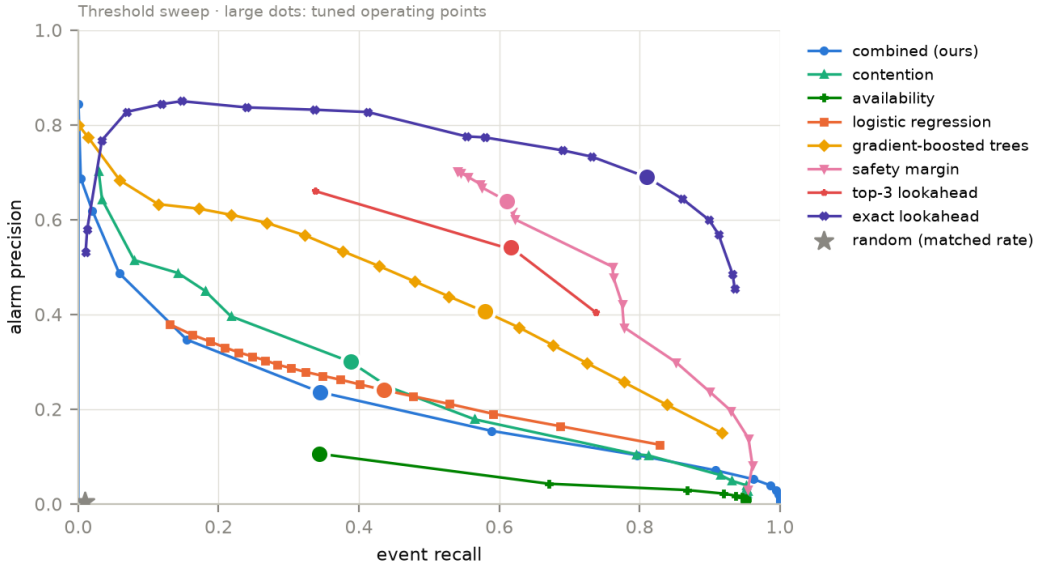


Figure 2: Precision–recall trade-off for every method (large dots: tuned settings; star: random).

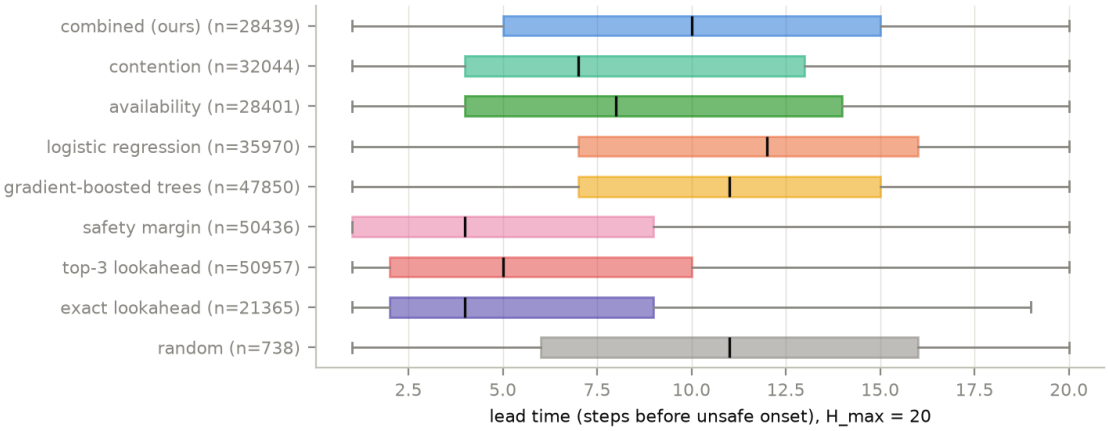


Figure 3: Lead times of the danger events each method caught.

Precision = share of alarms that were followed by danger within 20 steps. *Recall* = share of dangers that were warned about in advance. *F1* combines both. Lookahead is computed on part of the runs (all runs up to 25 processes, a sample above), so on exactly those runs: lookahead F1 = 0.74, combined = 0.44.

Reading it: our score is far better than chance and clearly better than the simple free-resource rule. On danger events that both caught, it warned earlier than the free-resource rule (11 vs 8 steps median); against lookahead the lead times were similar (5 vs 4 steps over 8,012 shared events).

6.3 Which signals matter (ablation)

Variant	F1
all five indicators, equal weights	0.24
without scarcity	0.26
without need pressure	0.25
without contention	0.20
without waiting ratio	0.27
without request rate	0.22

Removing contention or request rate hurts the equal-weight combination most; removing waiting ratio *helps*. Adding weak signals to a strong one dilutes it, which is why the tuner's best combination doubled contention's weight.

6.4 Machine learning, the safety margin and the cost of being exact

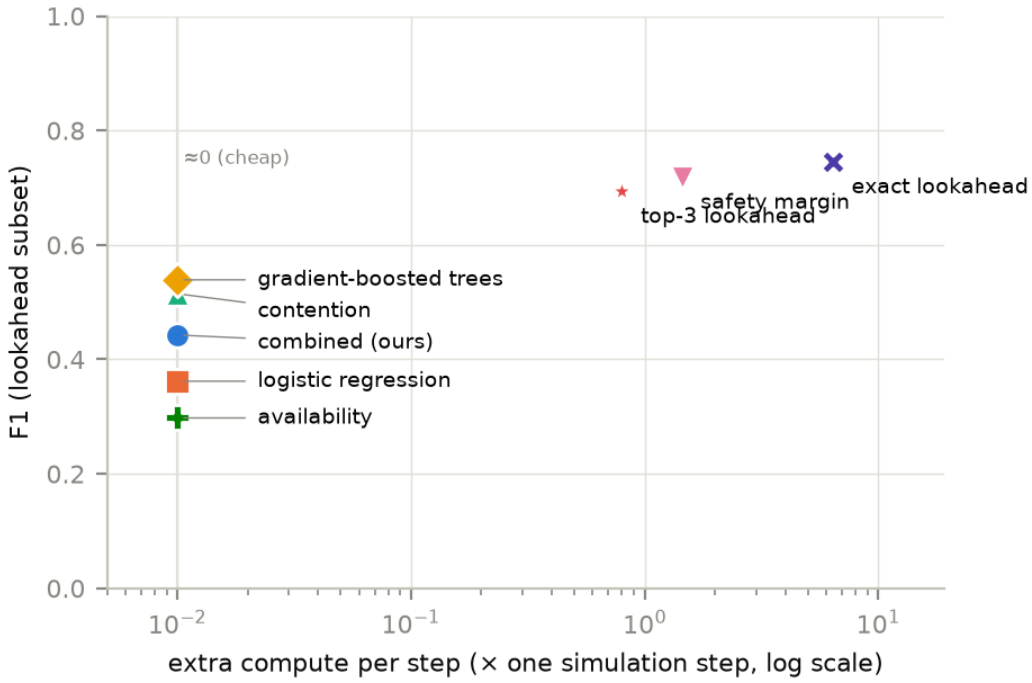


Figure 4: Accuracy against compute: cheap signals (left) vs checks that test the safety boundary directly (right).

Computation	Processes	× one simulation step
Safety margin	all	1.44
Safety margin	200	1.66
Top-3 lookahead	all	0.80
Top-3 lookahead	200	0.91
Exact lookahead (expensive)	all	6.40
Exact lookahead (expensive)	200	43.45

The learned models and the five signals cost almost nothing per step. The safety margin and top-3 lookahead use a fixed handful of safety checks, so their cost barely grows with system size, while exact lookahead runs one check per process and becomes very expensive for large systems.

Wrong maximum claims. When the signals were computed from claims that were off by up to ±20%, F1 changed by less than half a percentage point for every method, so the method does not need perfectly known claims.

Method	F1, true claims	F1, claims ±20%
Combined score (ours)	0.28	0.28
Need pressure	0.19	0.19
Contention	0.34	0.34
Combined, max	0.34	0.34
Logistic regression (ML)	0.31	0.31
Gradient-boosted trees (ML)	0.48	0.48

6.5 By scenario (F1)

Method	normal	spike	deadlock_prone
Combined score (ours)	0.22	0.42	0.23
Contention	0.21	0.49	0.22
Need pressure	0.06	0.34	0.01
Waiting ratio	0.29	0.03	0.31
Request rate	0.00	0.23	0.00
Availability (free-resource rule)	0.10	0.34	0.05
Logistic regression (ML)	0.22	0.44	0.23
Gradient-boosted trees (ML)	0.36	0.58	0.43
Safety margin	0.55	0.78	0.54
Top-3 lookahead	0.45	0.80	0.44
Exact lookahead (expensive)	0.66	0.83	0.69
Random alarms (chance)	0.00	0.01	0.01

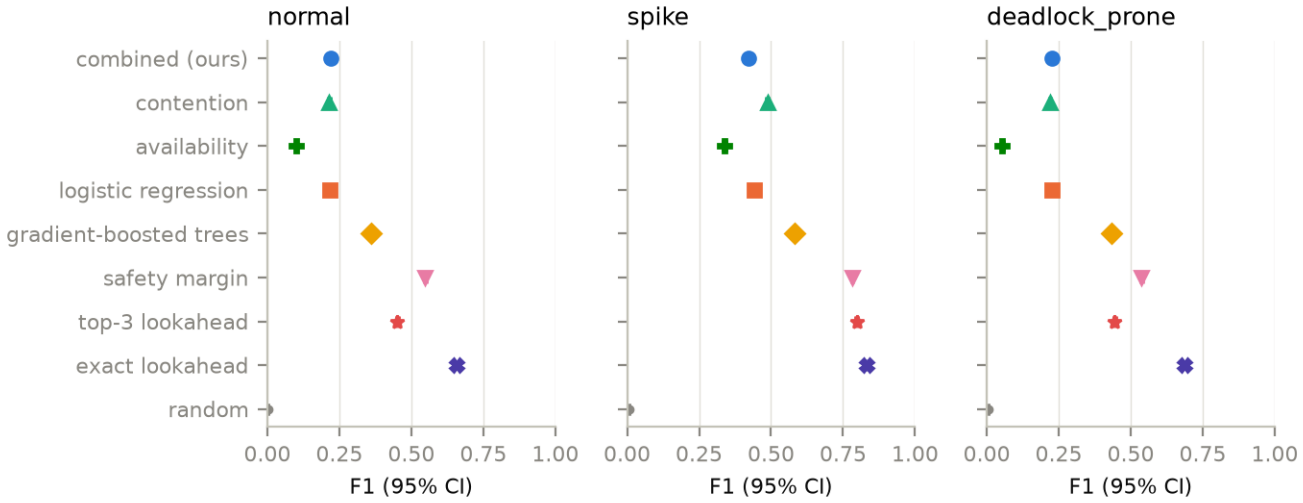


Figure 5: F1 per scenario with 95% confidence intervals.

Spikes are easiest: several signals rise together, and the danger starts a median 26 steps after the burst begins (only 3.0% before it). Slow build-ups give little warning before the line is crossed; there, waiting ratio alone does better than the combination. Contention's overall lead comes mainly from spikes.

6.6 Unsafe versus deadlock

Scenario	Recovered	Became deadlock	Deadlock share
deadlock_prone	0	31219	100%

Scenario	Recovered	Became deadlock	Deadlock share
normal	4912	16613	77%
spike	0	29868	100%

Predicting the deadlock itself (alarms allowed while already unsafe):

Method	F1 (H=20)	F1 (H=50)
Combined score (ours)	0.33	0.53
Contention	0.26	0.44
Need pressure	0.23	0.36
Waiting ratio	0.48	0.65
Request rate	0.06	0.13
Availability (free-resource rule)	0.16	0.29
Logistic regression (ML)	0.29	0.49
Gradient-boosted trees (ML)	0.48	0.69
Safety margin	0.16	0.40
Top-3 lookahead	0.13	0.35
Exact lookahead (expensive)	0.37	0.76
Random alarms (chance)	0.01	0.02

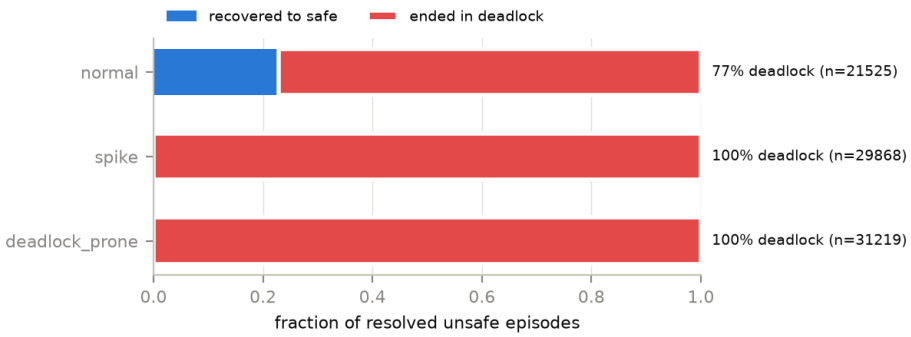


Figure 6: What happened to unsafe episodes in each scenario.

With a 20-step window, waiting ratio is the best deadlock predictor (F1 0.48): blocked processes pile up as a deadlock forms. With a 50-step window lookahead leads (0.76, on its subset of runs).

6.7 Warning window

With the same alarms, a wider window counts more alarms as correct: combined F1 = 0.11 (5 steps), 0.19 (10), 0.28 (20), 0.40 (50).

6.8 Speed

The NumPy (vectorised) Banker's check was **1.4× to 7.7× faster** than the plain textbook loop (from 10 to 200 processes). Theory says the check costs $\Theta(m \cdot n^2)$ in the worst case, but the measured growth was much gentler (log–log slope ≈ 0.27 vectorised and 0.76 for the textbook loop at $m = 20$): real states usually clear in a few passes, so the worst case almost never happens.

Figures for all of these are in `paper/figures/` (timelines, lead times, precision–recall curves, per-scenario F1, window sensitivity, episode outcomes, scaling, speed-up).

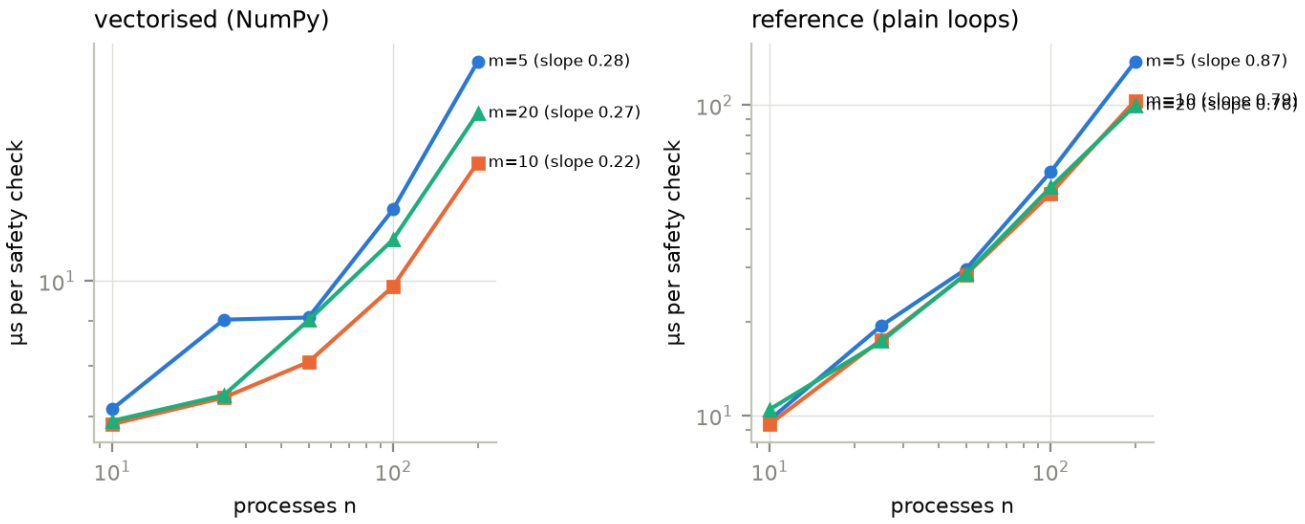


Figure 7: Time per Banker's safety check as the number of processes grows (log–log).

7. Limitations (honest)

- Workloads are synthetic, and processes must declare maximum claims (as Banker's algorithm requires); real systems rarely know them.
- The set of processes is fixed (no new arrivals); a run stops at the first deadlock.
- Claim tightness, which largely decides whether a run becomes unsafe, was chosen to give a balanced mix.
- Signals and Banker's check read the same state, so some correlation is built in — this is why we compare against random, free-resource and lookahead baselines.
- Lookahead was computed on a subset of runs for large systems.
- Three single-signal baselines (need pressure, request rate, availability) chose the highest threshold in the tuning grid, so slightly better settings may exist for them.

8. How to reproduce

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -m pytest
TUNING_CFG=configs/b_tuning.yaml MAIN_CFG=configs/b_main.yaml \
TUNING_RUN=results/b_tuning MAIN_RUN=results/b_main ./run_all.sh
```

AWS launch, monitoring and clean-up scripts are in `deployment/` (`aws_launch.sh`, `aws_watch.sh`, `aws_teardown.sh`, `aws_audit.sh`).

9. Presentation talking points

1. Banker's says "safe now". We ask: can we see trouble coming?
2. Show Figure 1: risk score rising, alarm, then unsafe and deadlock a few steps later.
3. No machine learning on purpose: explainable, and a test of whether simple signals are enough.
4. Correctness: safe never coexists with deadlock; unsafe can recover; lead time 0 is a miss; long alarms count once; alarms at the end of a cut-off run are excluded.
5. Results: F1 0.28 vs 0.01 random and 0.16 free-resource rule; median 10 steps of warning; boosted trees on the same signals 0.48; safety margin 0.72 at a fraction of exact lookahead's cost (0.74, 43×).
6. Scale: 195,000 sequences, 440 million safety checks, run on AWS EC2.
7. Limitations and future work: real traces, learned models, two-stage warning (cheap score first, lookahead only when the score is high).